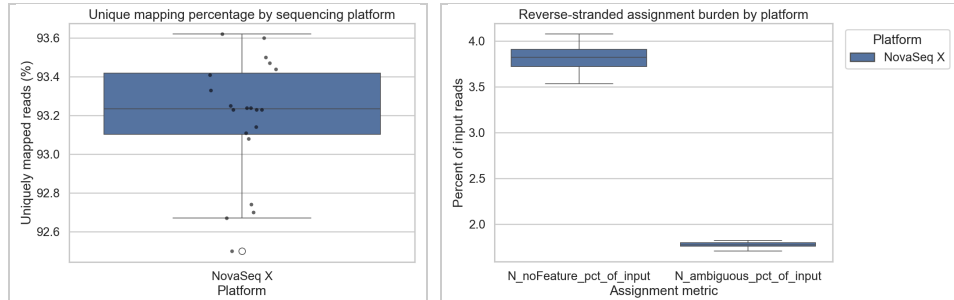


What I accomplished since the previous report

Since the previous report, we shifted from exporting DE results to interpreting the `mouse_new` contingency dataset. The goal was to determine which signals are strong enough to support the next report and prepare for Draft 1. Using the current SRP618841 notebook and derived outputs, we reviewed dataset structure, re-plotted the main contrasts, and added follow-up analyses to make the results easier to interpret.

The clearest conclusion is that the side-specific DRG contrasts remain the strongest and cleanest signal in the dataset. `geno_in_contra` is the strongest secondary branch, whereas `geno_in_ipsi` and `interaction` remain exploratory.



Alignment checkpoint (STAR, fastp-clean; pre-DE)

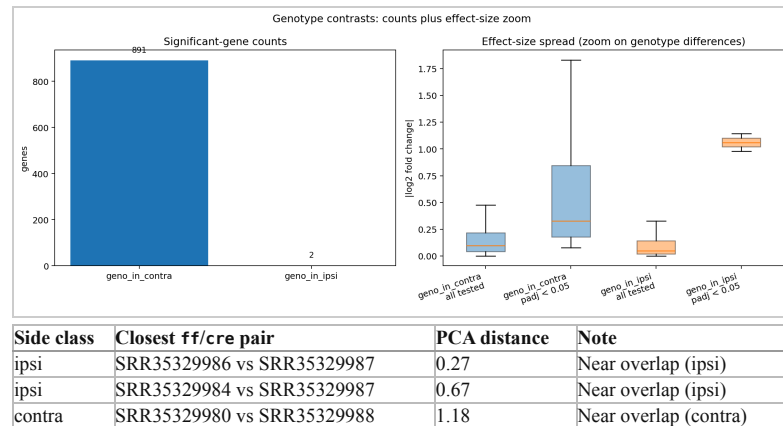
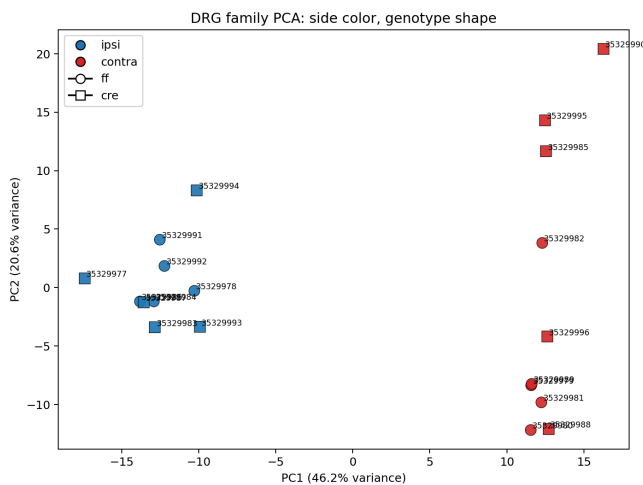
Left plot (unique mapping by platform): median unique mapping is **93.24%** (IQR **93.10–93.42%**), with median multi-mapping **5.10%**. Right plot (assignment burden by platform): reverse-stranded assignment burden is low (**~3.82%** noFeature, **~1.77%** ambiguous), so downstream PCA/DE structure is interpreted primarily as contrast/biological signal.

The bend-point cutoff is a narrowing rule derived from the raw ordered-*p*-value cumulative curve. It is used to choose downstream gene sets for enrichment and story selection when a simple $p_{adj} \leq 0.05$ cutoff yields an impractically large list.

Following class discussion, I used a PCA-first reading and a reproducible bend-point (elbow) threshold on the ordered-*p*-value cumulative curve to narrow large DE lists into smaller downstream sets. The thresholds and resulting gene-set sizes are summarized here to help the team choose the main story and supporting context.

Selected outputs for this report: The alignment checkpoint above plus Figures 1–4 show the PCA-first dataset structure, bend-point narrowing for the main side-specific contrast, the supporting genotype branch, and enrichment used as interpretation support.

Figure 1. PCA-first reading of the `mouse_new` DRG family



This PCA is the correct first read because it shows baseline structure (and collisions) before any gene-level filtering, preventing over-interpretation of long DE lists when groups do not separate cleanly. In Figure 1, the annotated PCA (left) separates more clearly by side than genotype, and the genotype count/effect-size zoom plus collision table (right) confirm near-overlapping ff/cre pairs; genotype is therefore treated as secondary in this report.

Figure 2. Main side-specific contrast after class-guided filtering

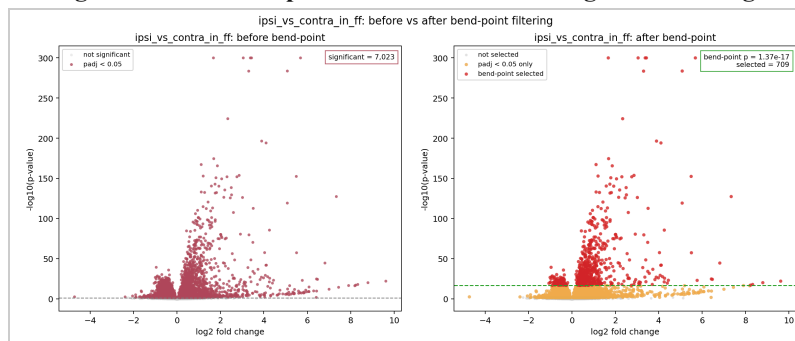


Figure 3. Secondary genotype branch: `geno_in_contra`

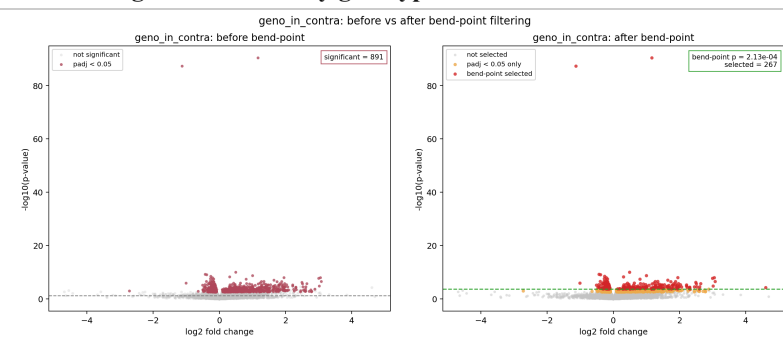
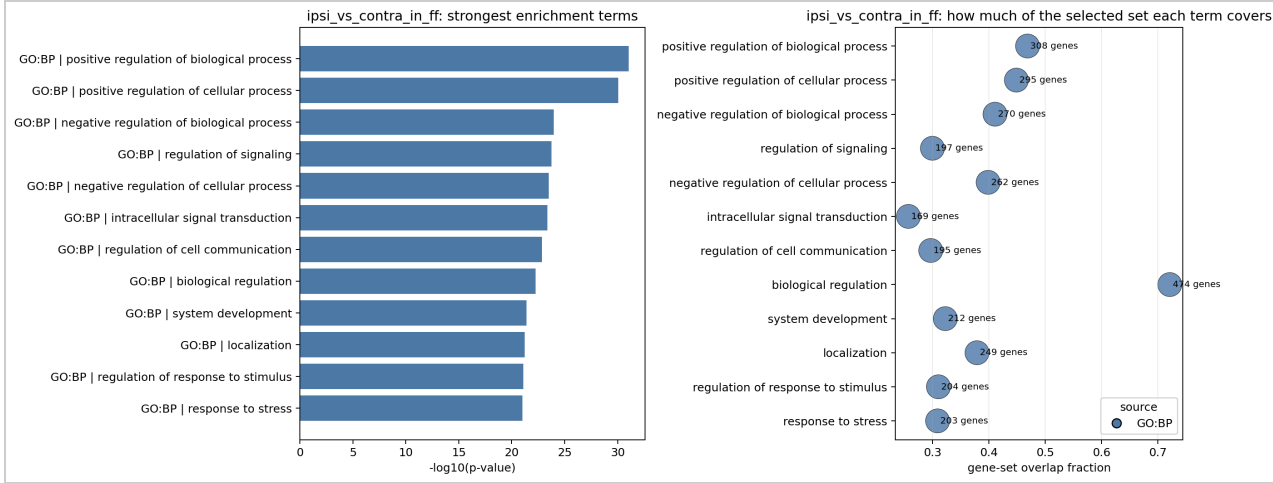


Figure 2 shows the main side-specific contrast (`ipsi_vs_contra_in_ff`) after bend-point narrowing and supports it as the primary story candidate without an arbitrary top-*N* cutoff. Counts: $p_{adj} \leq 0.05$ yields 7,023 genes, and the bend-point selects a 709-gene core (6,314 p_{adj} -only). Figure 3 shows the strongest genotype-specific supporting branch (`geno_in_contra`) after the same step: 891 ($p_{adj} \leq 0.05$) → 267 core (631 p_{adj} -only). It is kept as context while the side-specific DE signal remains central.



Methods used

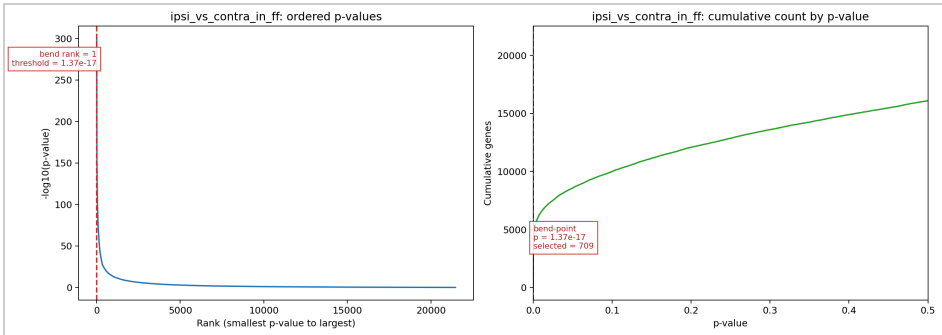
This report keeps the same DESeq2 result framework from previous reports, but the emphasis is now interpretive rather than purely procedural. We treated PCA as the first check, compared signal strength across contrasts, and then applied an ordered-p-value / cumulative bend-point method to narrow large gene lists into smaller downstream sets for enrichment and story selection. The ordered-p-value cumulative curve shown at right (for `ipsi_vs_contra_in_ff`) is the diagnostic used to set the bend-point cutoff.

```
Repro (from Semester5/BIOL550/group_project/):
jupyter nbconvert --execute --to notebook --inplace \
  mouse_new/notebooks/mouse_differential_expression_all20.ipynb

Canonical notebook + derived outputs:
- notebook:
  mouse_new/notebooks/mouse_differential_expression_all20.ipynb
- derived outputs:
  mouse_new/differential_expression_all20/derived_analysis/
- signal summary:
  mouse_new/differential_expression_all20/derived_analysis/analysis_summary.tsv

Family + DESeq2 model (from the family manifest):
- family: family_drg_novaseqx (20 samples)
- design: ~ side_class + geno_class + side_class:geno_class

Contrasts reviewed in this report:
- ipsi_vs_contra_in_ff
- ipsi_vs_contra_in_cre
- geno_in_contra
- geno_in_ipsi
- interaction
```



- PCA is used first to separate intrinsic sample structure from contrast-specific signal.
- Bend-point filtering is used to define a narrower core set from the much larger significant-gene pool.
- Enrichment (GO / KEGG / Reactome via g:Profiler) is used to interpret the bend-point-selected sets after the main contrast is understood.

Problems encountered

The main challenge this week was deciding how to interpret it without forcing a misleading story. The raw significant-gene counts were large enough to be hard to use directly, and the genotype signal was easy to over-read if we looked at labels without grounding them in the PCA structure first. We also deciding which figures actually explained the data and which ones simply repeated the same point in a different format.

We have addressed some of this by using PCA-first interpretation, bend-point-based narrowing, and more selective figure choices. This keeps the side-specific DRG contrasts central, retains `geno_in_contra` as supporting context, and treats weaker genotype/interaction branches as exploratory.

Goals for the coming week

Next week the main goal is to turn this DE analysis into a cleaner starting point for draft 1 outlining. The strongest claim candidate is now clearer: the side-specific DRG signal is the main story, and the best supporting branch is `geno_in_contra`. We should keep using the PCA-first and bend-point-guided structure so the report and the paper stay aligned.

- Choose which side-specific contrast should be the main figure for the next weekly report and the paper draft1.
- Keep `geno_in_contra` as the main supporting genotype branch and avoid over-centering `geno_in_ipsi` OR `interaction`.
- Use the bend-point-selected gene sets and enrichment companion plots to choose the clearest biological themes for the paper draft1.
- Carry this structure directly into the paper outline so the report and the paper stay aligned.